

Diabetes Classification & Clustering

Multi-Class Prediction of Diabetes Status

Classes: N = Non-diabetic | Y = Diabetic | P = Predict-diabetic

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0 . Imports and Configuration

All imports successful.

1 . Data Loading and Initial Inspection

Shape: (1001, 14)
Columns: ['ID', 'No_Pation', 'Gender', 'AGE', 'Urea', 'Cr', 'HbA1c', 'Chol', 'TG', 'HDL', 'LDL', 'VLDL', 'BMI', 'CLASS']

	ID	No_Pation	Gender	AGE	Urea	Cr	HbA 1 c	Chol	TG	HDL	
0	5 0 2	1 7 9 7 5.0	F	5 0.0	4.7	4 6.0	4.9	4.2	0.9	2.4	1
1	7 3 5	3 4 2 2 1.0	M	2 6.0	4.5	6 2.0	4.9	3.7	1.4	1.1	2
2	4 2 0	4 7 9 7 5.0	F	5 0.0	4.7	4 6.0	4.9	4.2	0.9	2.4	1
3	6 8 0	8 7 6 5 6.0	F	5 0.0	4.7	4 6.0	4.9	4.2	0.9	2.4	1
4	5 0 4	3 4 2 2 3.0	M	3 3.0	7.1	4 6.0	4.9	4.9	1.0	0.8	2
5	6 3 4	3 4 2 2 4.0	F	4 5.0	2.3	2 4.0	4.0	2.9	1.0	1.0	1
6	7 2 1	3 4 2 2 5.0	F	5 0.0	2.0	5 0.0	4.0	3.6	1.3	0.9	2
7	4 2 1	3 4 2 2 7.0	M	4 8.0	4.7	4 7.0	4.0	2.9	0.8	0.9	1
8	6 7 0	3 4 2 2 9.0	M	4 3.0	2.6	6 7.0	4.0	3.8	0.9	2.4	3
9	7 5 9	3 4 2 3 0.0	F	3 2.0	3.6	2 8.0	4.0	3.8	2.0	2.4	3

<class 'pandas.DataFrame'>

RangeIndex: 1001 entries, 0 to 1000

Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	ID	1001 non-null	str
1	No_Pation	1000 non-null	float64
2	Gender	1000 non-null	str
3	AGE	1000 non-null	float64
4	Urea	1000 non-null	float64
5	Cr	1000 non-null	float64
6	HbA1c	1000 non-null	float64
7	Chol	1000 non-null	float64
8	TG	1000 non-null	float64
9	HDL	1000 non-null	float64
10	LDL	1000 non-null	float64
11	VLDL	1000 non-null	float64
12	BMI	1000 non-null	float64
13	CLASS	1000 non-null	str

dtypes: float64(11), str(3)

memory usage: 109.6 KB

Missing values per column:

```
No_Pation    1
Gender       1
AGE          1
Urea         1
Cr           1
HbA1c        1
Chol         1
TG           1
HDL          1
LDL          1
VLDL         1
BMI          1
CLASS        1
dtype: int64
```

Dropping 1 row(s) with missing values...

Shape after dropping NAs: (1000, 14)

Shape after dropping IDs: (1000, 12)

	Gender	AGE	Urea	Cr	HbA 1 c	Chol	TG	HDL	LDL	VLDL	BMI	CLA
0	F	50.0	4.7	46.0	4.9	4.2	0.9	2.4	1.4	0.5	24.0	
1	M	26.0	4.5	62.0	4.9	3.7	1.4	1.1	2.1	0.6	23.0	
2	F	50.0	4.7	46.0	4.9	4.2	0.9	2.4	1.4	0.5	24.0	
3	F	50.0	4.7	46.0	4.9	4.2	0.9	2.4	1.4	0.5	24.0	
4	M	33.0	7.1	46.0	4.9	4.9	1.0	0.8	2.0	0.4	21.0	

Unique Gender values: <StringArray>

['F', 'M']

Length: 2, dtype: str

Unique CLASS values: <StringArray>

['N', 'P', 'Y']

Length: 3, dtype: str

Gender distribution:

Gender

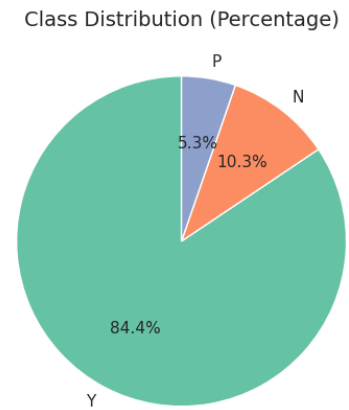
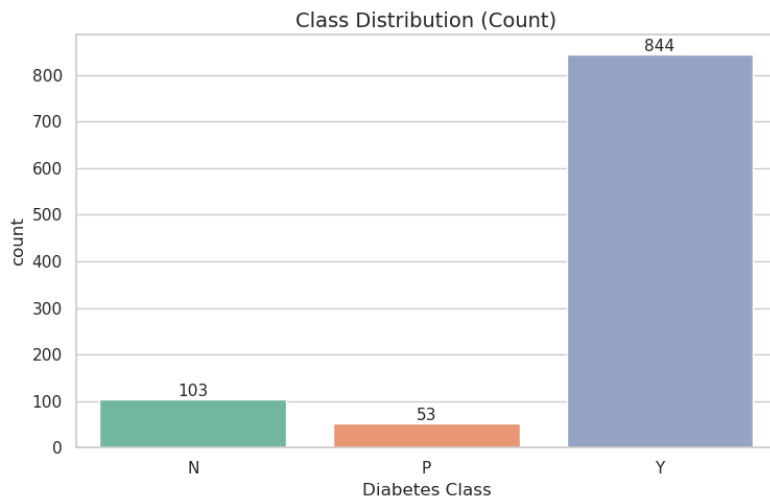
M 565

F 435

Name: count, dtype: int64

2. Exploratory Data Analysis — Univariate

2.1 Target Variable Distribution



Class Distribution Summary:

	Count	Percentage
CLASS		
Y	844	84.4
N	103	10.3
P	53	5.3

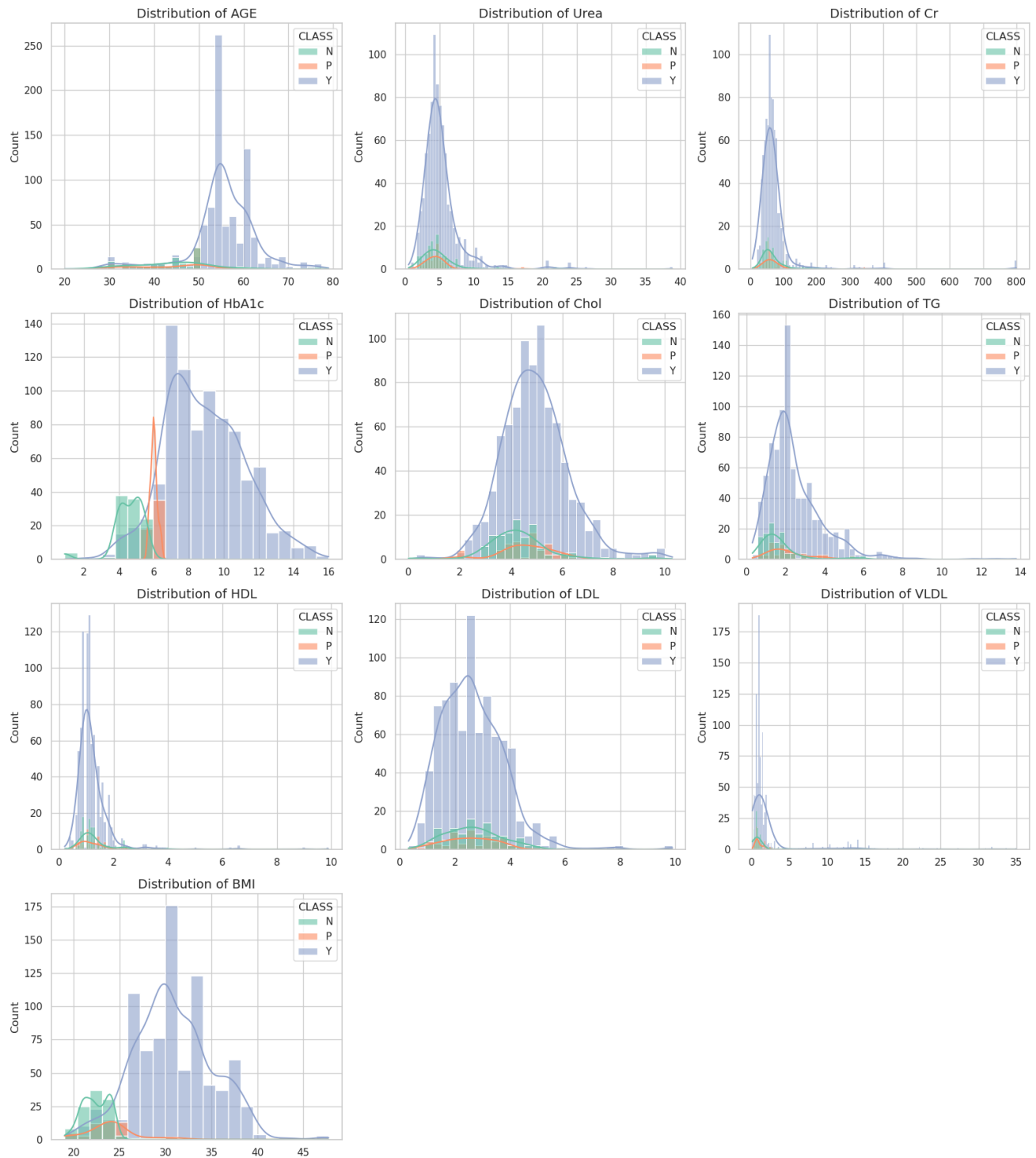
Imbalance ratio (largest/smallest): 15.9:1

2.2 Numeric Feature Summary Statistics

Numeric features (10): ['AGE', 'Urea', 'Cr', 'HbA1c', 'Chol', 'TG', 'HDL', 'LDL', 'VLDL', 'BMI']

	count	mean	std	min	2 5 %	5 0 %	7 5 %
AGE	1 0 0 0 . 0	5 3 . 5 2 8 0 0 0	8 . 7 9 9 2 4 1	2 0 . 0	5 1 . 0	5 5 . 0	5 9 . 0
Urea	1 0 0 0 . 0	5 . 1 2 4 7 4 3	2 . 9 3 5 1 6 5	0 . 5	3 . 7	4 . 6	5 . 9
Cr	1 0 0 0 . 0	6 8 . 9 4 3 0 0 0	5 9 . 9 8 4 7 4 7	6 . 0	4 8 . 0	6 0 . 0	7 1 . 0
HbA 1 c	1 0 0 0 . 0	8 . 2 8 1 1 6 0	2 . 5 3 4 0 0 3	0 . 9	6 . 5	8 . 0	1 0 . 0
Chol	1 0 0 0 . 0	4 . 8 6 2 8 2 0	1 . 3 0 1 7 3 8	0 . 0	4 . 0	4 . 8	5 . 6
TG	1 0 0 0 . 0	2 . 3 4 9 6 1 0	1 . 4 0 1 1 7 6	0 . 3	1 . 5	2 . 0	2 . 7
HDL	1 0 0 0 . 0	1 . 2 0 4 7 5 0	0 . 6 6 0 4 1 4	0 . 2	0 . 9	1 . 1	1 . 5
LDL	1 0 0 0 . 0	2 . 6 0 9 7 9 0	1 . 1 1 5 1 0 2	0 . 3	1 . 8	2 . 5	3 . 0
VLDL	1 0 0 0 . 0	1 . 8 5 4 7 0 0	3 . 6 6 3 5 9 9	0 . 1	0 . 7	0 . 9	1 . 2
BMI	1 0 0 0 . 0	2 9 . 5 7 8 0 2 0	4 . 9 6 2 3 8 8	1 9 . 0	2 6 . 0	3 0 . 0	3 3 . 0

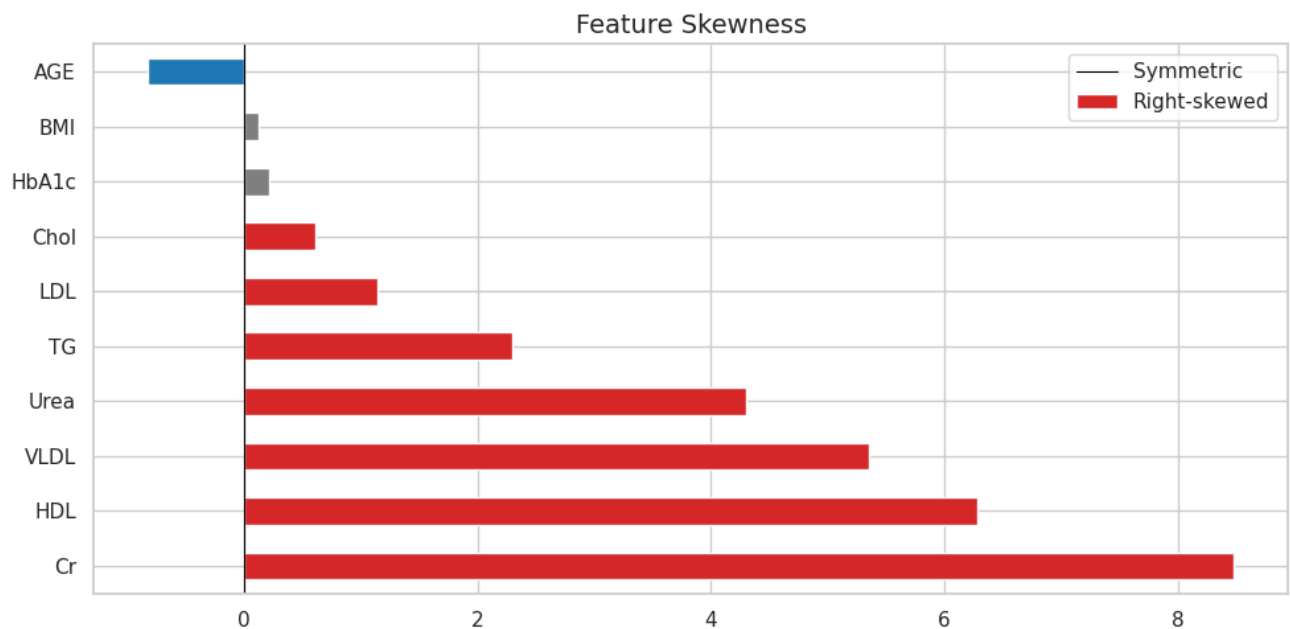
2.3 Feature Distributions — Histograms with KDE



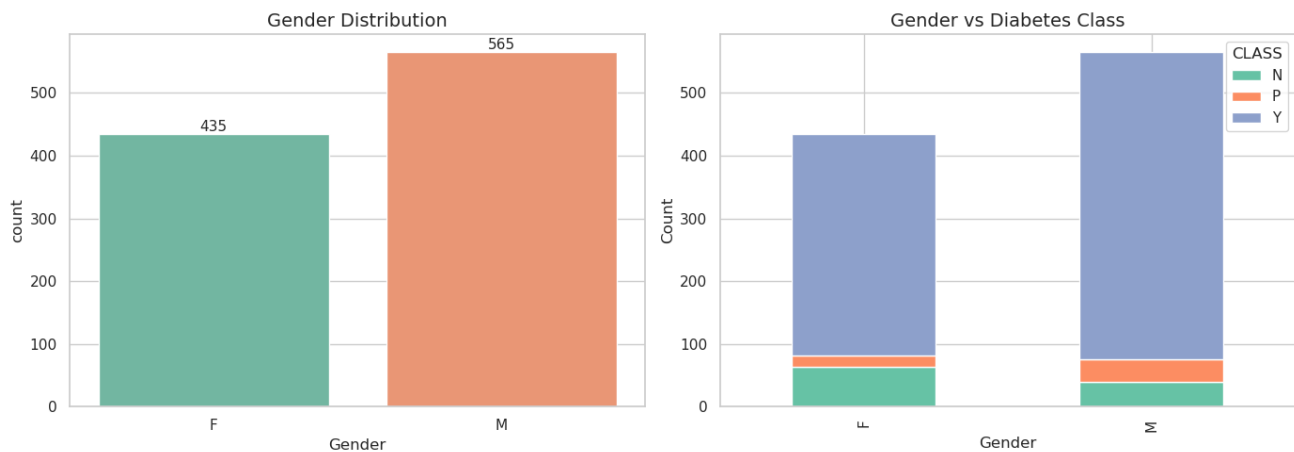
2.4 Skewness Analysis

Skewness of numeric features:

Cr	:	+8.474	(right-skewed)
HDL	:	+6.283	(right-skewed)
VLDL	:	+5.350	(right-skewed)
Urea	:	+4.299	(right-skewed)
TG	:	+2.298	(right-skewed)
LDL	:	+1.146	(right-skewed)
Chol	:	+0.617	(right-skewed)
HbA1c	:	+0.222	(symmetric-skewed)
BMI	:	+0.126	(symmetric-skewed)
AGE	:	-0.820	(left-skewed)



2 . 5 Gender Distribution



Gender x CLASS Crosstab:

CLASS	N	P	Y
Gender			
F	64	17	354
M	39	36	490

Gender proportions:

Gender

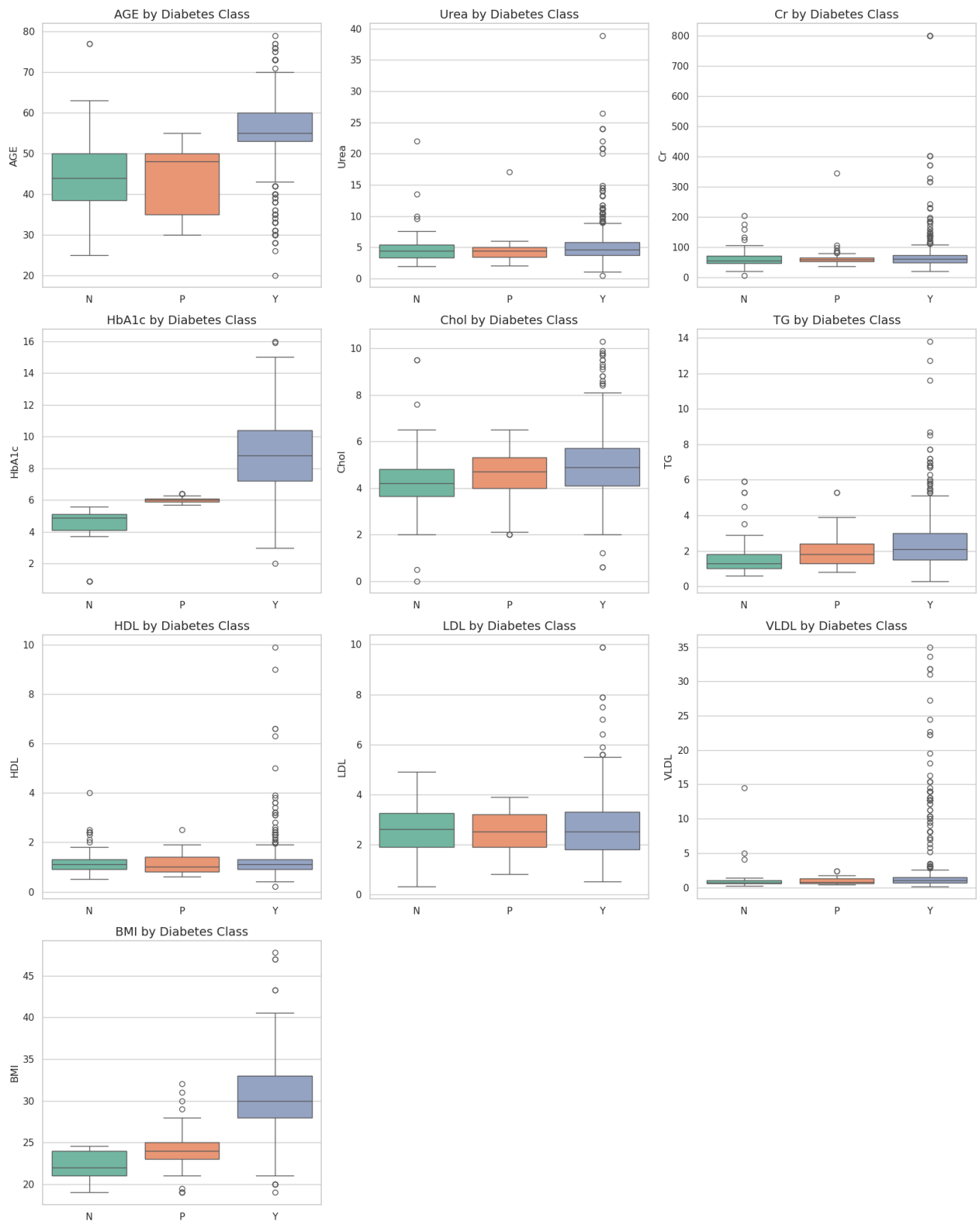
M 56.5

F 43.5

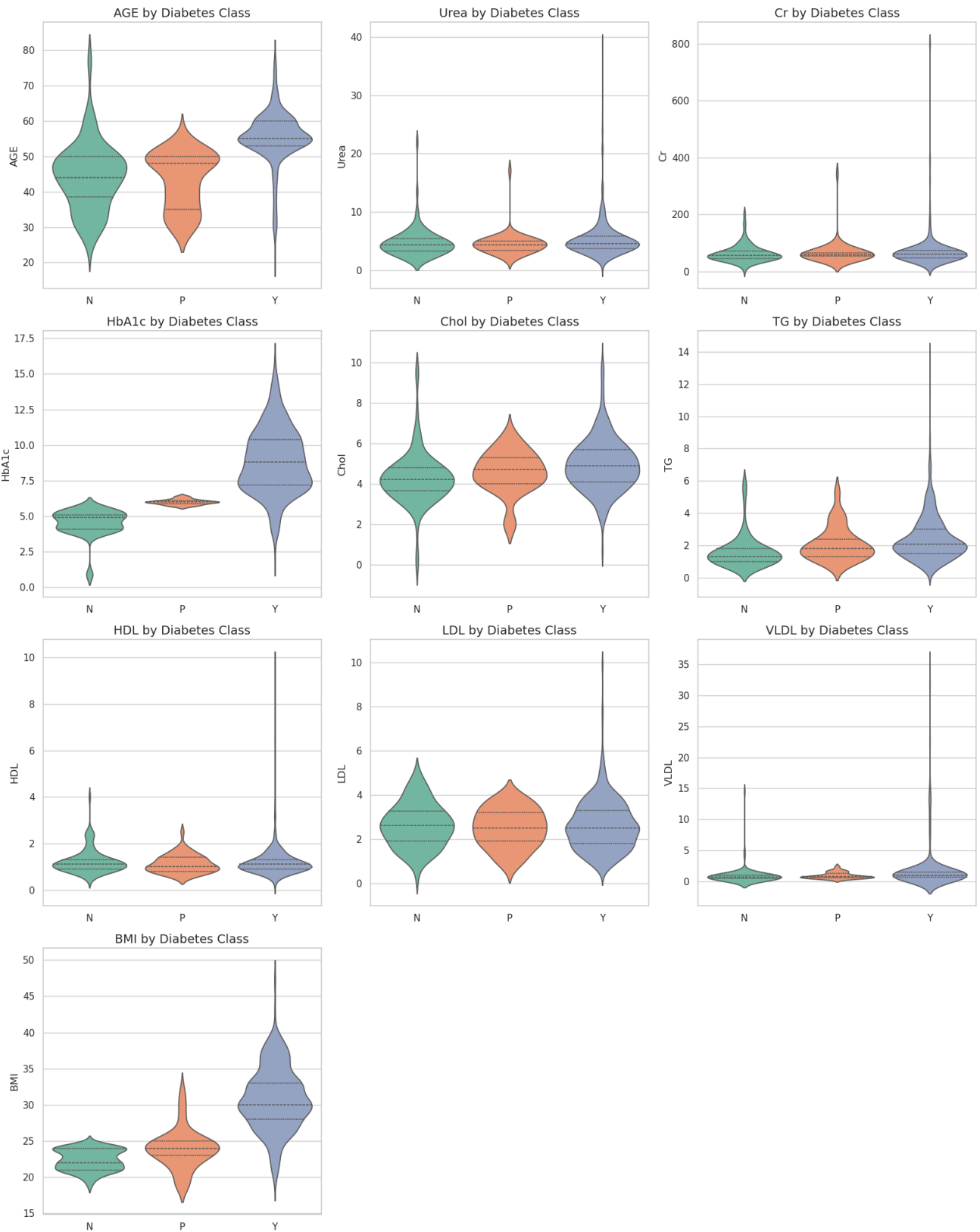
Name: proportion, dtype: float64

3 . Exploratory Data Analysis — Bivariate

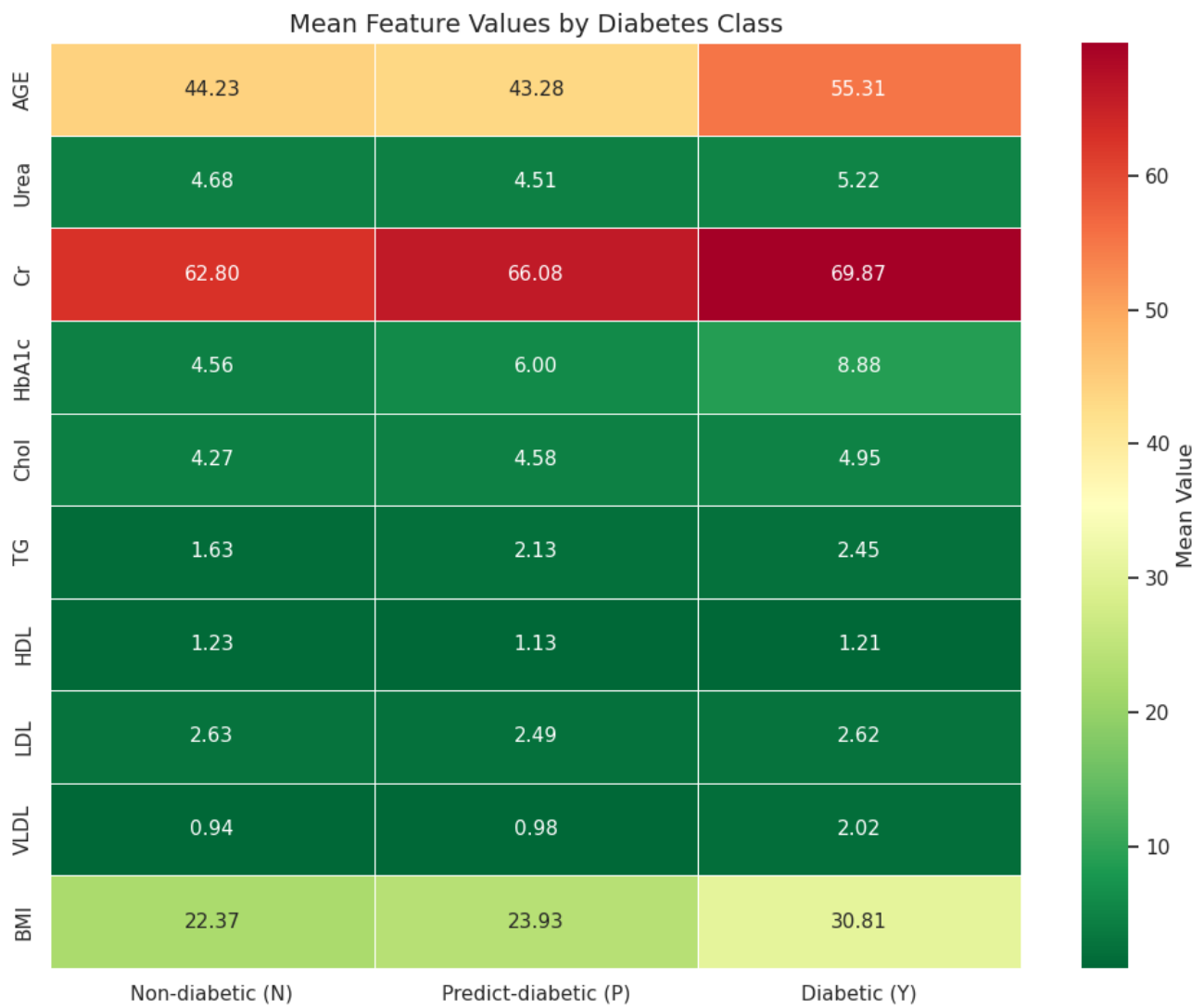
3 . 1 Feature vs Class — Boxplots



3.2 Feature vs Class — Violin Plots

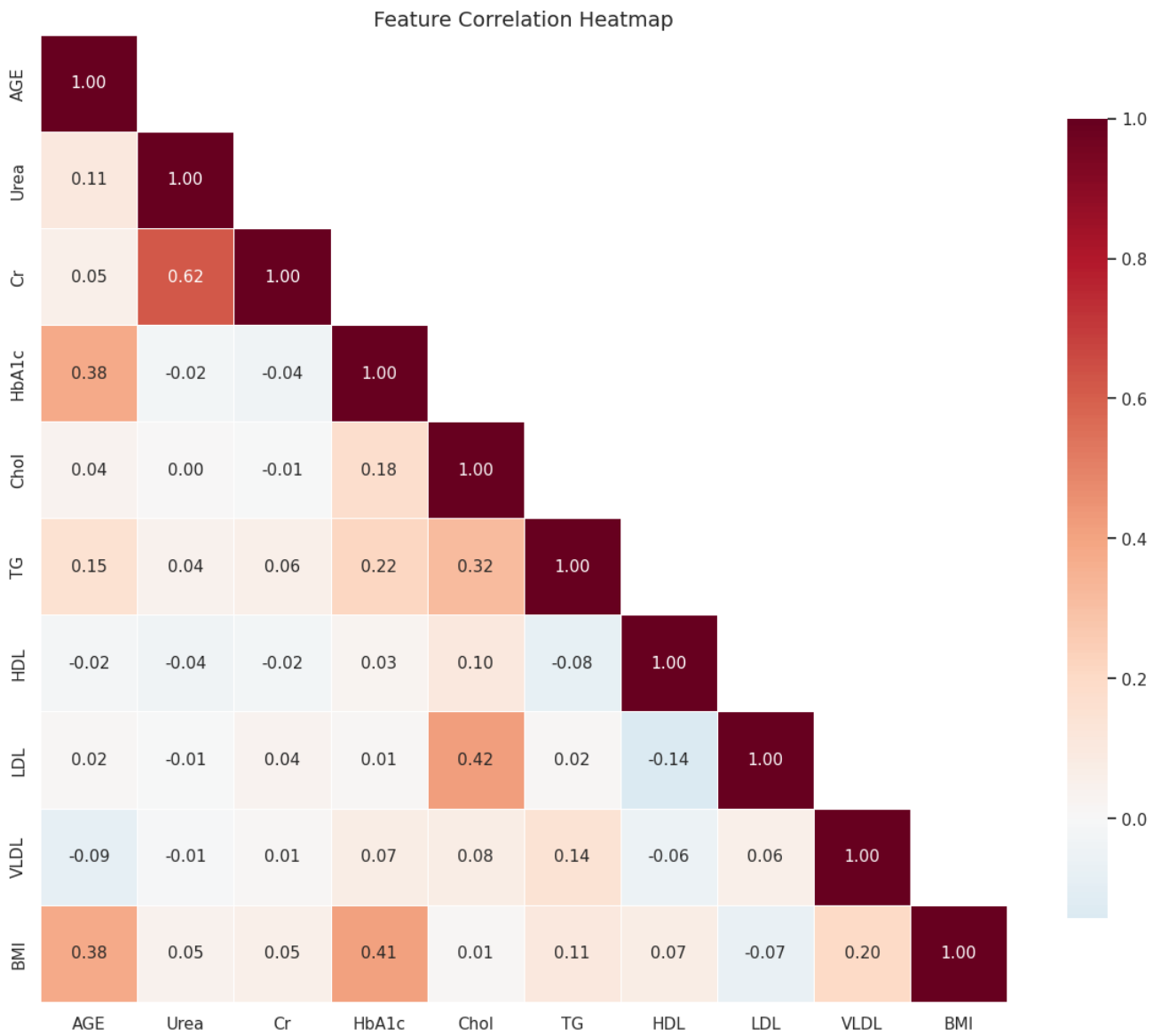


3.3 Class-wise Mean Comparison



4 . Exploratory Data Analysis — Multivariate

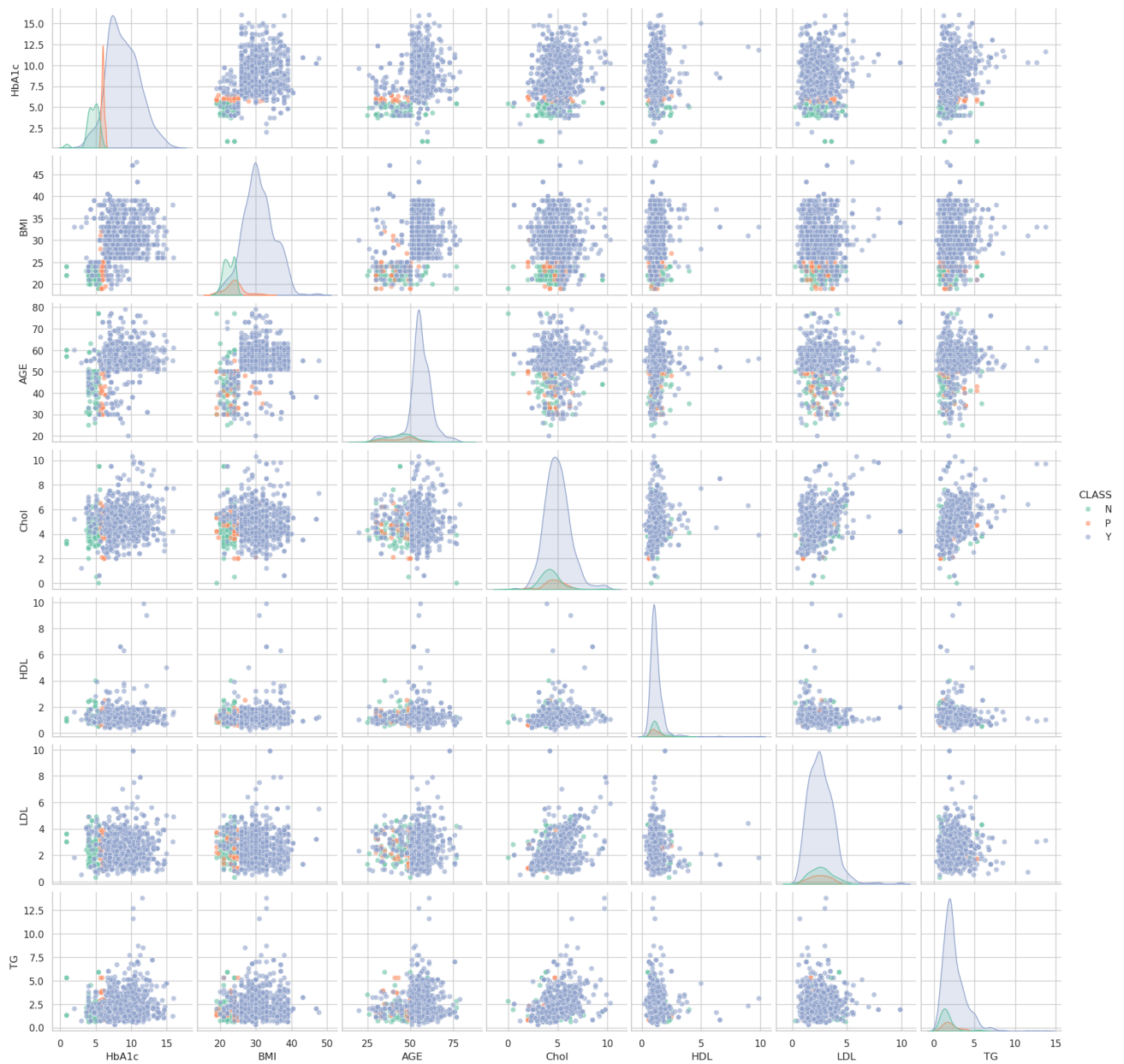
4 . 1 Correlation Heatmap



Strong feature correlations ($|r| > 0.7$):
None found.

4 . 2 Pairplot of Key Features

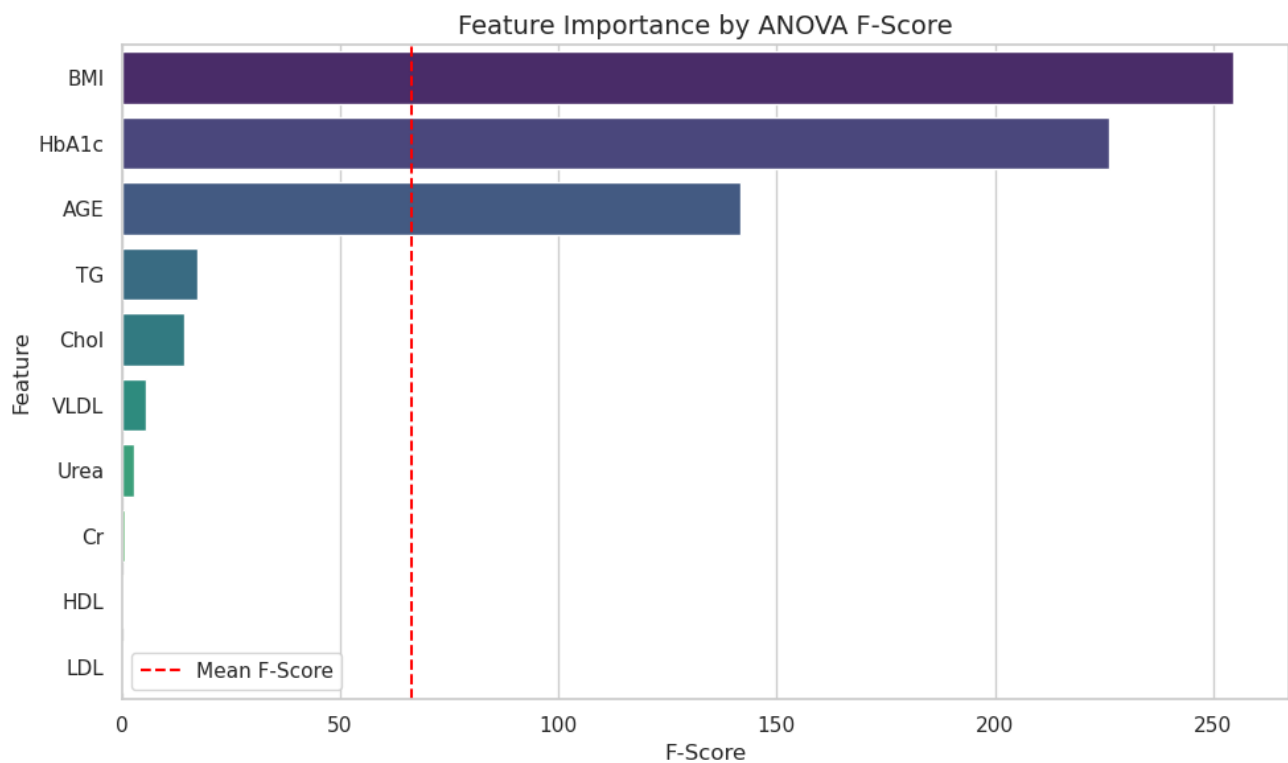
Pairplot of Key Features by Diabetes Class



4 . 3 ANOVA F-Test — Feature Discriminative Power

ANOVA F-Test Results (ranking features by discriminative power):

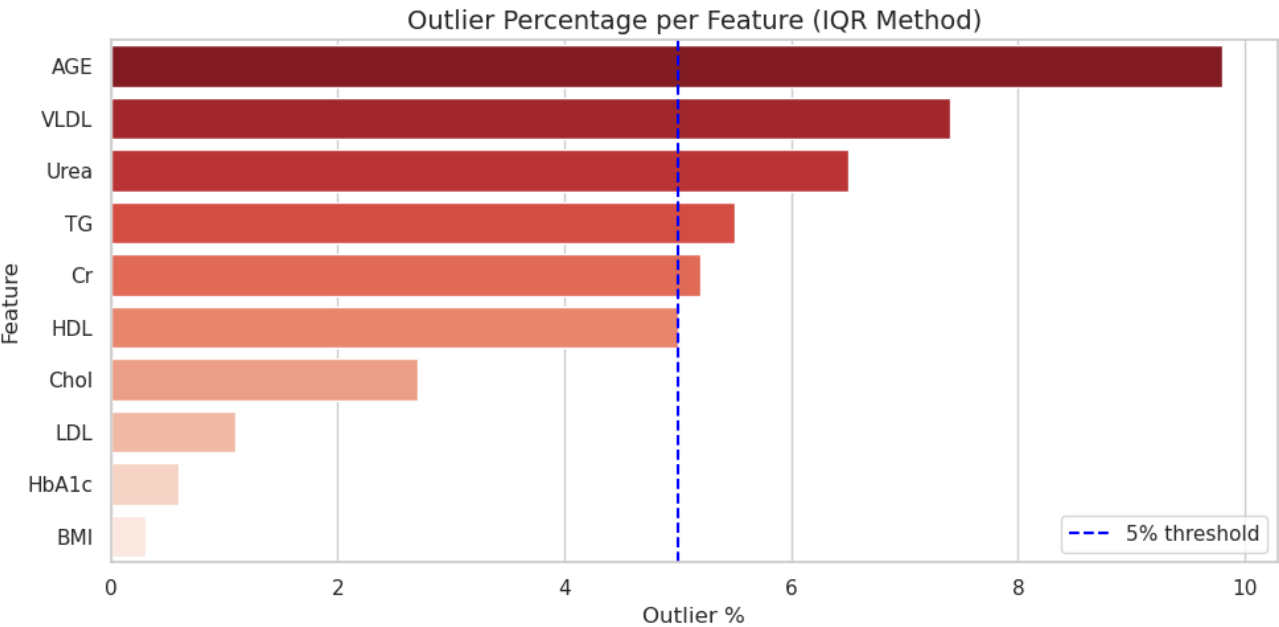
Feature		F-Score										p-value									
9	BMI	2	5	4	.6	0	5	6	3	3	4	.7	1	4	3	2	3	e-9	0		
3	HbA1c	2	2	6	.1	4	5	6	1	3	1	.0	3	1	4	2	7	e-8	1		
0	AGE	1	4	1	.7	9	2	1	8	2	6	.4	0	5	4	7	7	e-5	5		
5	TG	1	7	.2	3	3	2	4	0		4	.3	8	6	9	2	0	e-0	8		
4	Chol	1	4	.3	8	2	1	8	2		6	.9	5	5	1	1	6	e-0	7		
8	VLDL		5	.6	0	1	3	0	5		3	.8	1	0	2	1	6	e-0	3		
1	Urea		2	.7	6	7	6	5	1		6	.3	2	9	2	0	2	e-0	2		
2	Cr		0	.7	0	2	4	2	5		4	.9	5	6	2	7	7	e-0	1		
6	HDL		0	.4	2	7	2	1	2		6	.5	2	4	4	4	5	e-0	1		
7	LDL		0	.3	0	8	1	6	4		7	.3	4	8	6	4	9	e-0	1		



5 . Outlier Detection and Analysis

5 . 1 IQR-Based Outlier Detection

	Feature	Q 1		Q 3		IQR	Lower Bound		Upper Bound		Outlier Count	Outlier %			
0	AGE	5	1.0	5	9.0	8.0	3	9.0	0	7	1.0	0	9.8		
8	VLDL	0	.7	1	.5	0.8	-	0.5	0	2	.7	0	7.4		
1	Urea	3	.7	5	.7	2.0	0	.7	0	8	.7	0	6.5		
5	TG	1	.5	2	.9	1.4	-	0.6	0	5	.0	0	5.5		
2	Cr	4	8.0	7	3.0	2	5.0	1	0.5	0	1	1	0.5	0	5.2
6	HDL	0	.9	1	.3	0.4	0	.3	0	1	.9	0	5.0		
4	Chol	4	.0	5	.6	1.6	1	.6	0	8	.0	0	2.7		
7	LDL	1	.8	3	.3	1.5	-	0.4	5	5	.5	5	1.1		
3	HbA 1 c	6	.5	1	0.2	3.7	0	.9	5	1	5.7	5	6	0.6	
9	BMI	2	6.0	3	3.0	7.0	1	5.5	0	4	3.5	0	3	0.3	



5.2 Domain-Specific Outlier Review

Clinical thresholds for flagging implausible values:

- **Cr (Creatinine):** Normal ~ 0.6 – 1.3 mg/dL; values > 2.0 suggest end-stage renal disease or data error
- **HbA 1 c:** Normal < 5.7 %; values < 3 % are physiologically implausible
- **Chol:** Values near 0 are likely data errors

Clinically implausible values flagged:

Cr > 200: 17 rows

AGE			Cr	HbA 1 c	Chol	BMI	CLASS
9 1	5 0.0	2 0 3.0		5.4	3.8	2 2.0	N
1 5 1	3 0.0	3 4 4.0		6.0	4.9	2 1.0	P
2 0 8	5 7.0	3 7 0.0		6.8	4.0	3 7.0	Y
2 1 2	5 7.0	3 7 0.0		6.8	6.1	3 7.0	Y
2 7 3	5 8.0	8 0 0.0		9.1	6.6	3 3.0	Y

HbA1c < 3: 5 rows

	AGE		Cr	HbA 1 c	Chol	BMI	CLASS
8 0	5 7.0	9 7.0		0.9	3.2	2 2.0	N
9 2	6 0.0	7 0.0		0.9	3.4	2 4.0	N
9 8	6 0.0	7 0.0		0.9	3.4	2 4.0	N
1 0 1	5 7.0	9 7.0		0.9	3.2	2 2.0	N
5 5 4	6 0.0	7 4.0		2.0	5.2	3 3.0	Y

Chol < 0.5: 1 rows

AGE		Cr	HbA 1 c	Chol	BMI	CLASS
9 9	7 7.0	1 0 6.0	5.4	0.0	1 9.0	N

5.3 Outlier Treatment Strategy

- **Tree-based models** (Random Forest, XGBoost) are robust to outliers — no capping needed
- **Linear models** (Logistic Regression) benefit from capped/transformed features
- **For linear models:** Apply `RobustScaler` or a `QuantileTransformer` as alternative to `StandardScaler`
- **For clinically implausible values:** Cap at reasonable bounds using domain knowledge
- **Moderate outliers:** Retain — they carry important clinical signal (sick patients have extreme values)

6. Data Preprocessing

6.1 Train-Test Split (Stratified)

Class mapping: {'N': 0, 'P': 1, 'Y': 2}
Target distribution: {'N': 103, 'P': 53, 'Y': 844}

Training set: 800 samples
Test set: 200 samples

Train class distribution:

N: 82 (10.2%)
P: 43 (5.4%)
Y: 675 (84.4%)

Test class distribution:

N: 21 (10.5%)
P: 10 (5.0%)
Y: 169 (84.5%)

6.2 Feature Encoding and Scaling

- **Gender:** Binary → Label Encode (F= 0 , M= 1)
- **Numeric features:** StandardScaler (z-score normalization)
- Combined with `ColumnTransformer`

Numeric features (10): ['AGE', 'Urea', 'Cr', 'HbA1c', 'Chol', 'TG', 'HDL', 'LDL', 'VLDL', 'BMI']

Categorical features (1): ['Gender']

Preprocessor: StandardScaler + OneHotEncoder(drop=first)

6.3 Handling Class Imbalance with SMOTE

The dataset is severely imbalanced (Y= 8 4 4 , N= 1 0 3 , P= 5 3). SMOTE generates synthetic samples for minority classes.

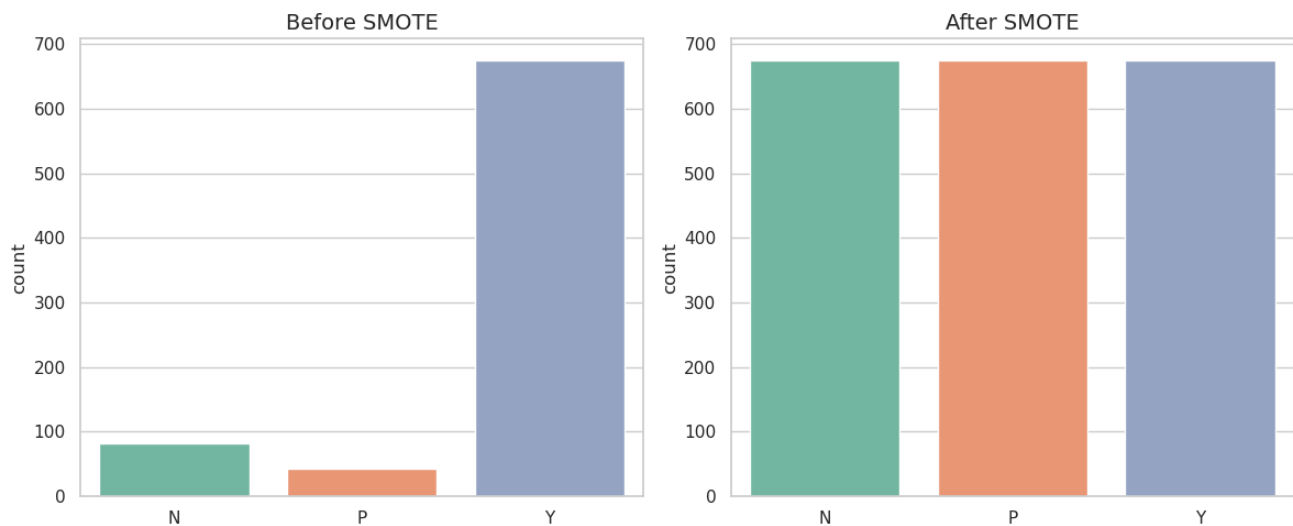
CRITICAL: SMOTE must be applied ONLY to training data and WITHIN cross-validation folds to prevent data leakage.

Class distribution BEFORE SMOTE:

N: 82
P: 43
Y: 675

Class distribution AFTER SMOTE:

N: 675
P: 675
Y: 675



6 . 4 Pipeline-Ready Preprocessor

For proper cross-validation, we will use `imblearn.pipeline.Pipeline` (not sklearn's) to chain:

- 1 . **ColumnTransformer** (scale numeric, encode categorical)
- 2 . **SMOTE** (oversample minority classes)
- 3 . **Classifier** (any sklearn-compatible model)

This ensures SMOTE is applied fold-by-fold during cross-validation, preventing data leakage.

7 . Classification — Baseline Models

7 . 1 Dummy Classifier (Lower Bound)

A model that always predicts the majority class (Y). Any useful model must beat this.

Dummy Classifier (most_frequent) – Test Accuracy: 0.8450

Dummy Classifier – Test F1-macro: 0.3053

Classification Report (Dummy):

	precision	recall	f1-score	support
N	0.00	0.00	0.00	21
P	0.00	0.00	0.00	10
Y	0.84	1.00	0.92	169
accuracy			0.84	200
macro avg	0.28	0.33	0.31	200
weighted avg	0.71	0.84	0.77	200

8 . Classification — Model Training with Cross-Validation

8 . 1 Model Definitions

Five models with imblearn pipelines (preprocessing + SMOTE + classifier):

Model	Rationale
Logistic Regression	Linear baseline, interpretable coefficients
Random Forest	Ensemble, robust to outliers, built-in feature importance
XGBoost	Gradient boosting, state-of-the-art for tabular data
SVC (RBF Kernel)	Margin-based, captures non-linear boundaries
K-Nearest Neighbors	Distance-based, non-parametric contrast

Pipelines defined for 5 models.

8 . 2 Stratified 5 -Fold Cross-Validation

Primary metric: **F 1 -macro** (treats all classes equally despite imbalance)

```
Running CV for Logistic Regression...
Running CV for Random Forest...
Running CV for XGBoost...
Running CV for SVC...
Running CV for KNN...
```

Cross-Validation Results (5-fold, mean scores):

	Accuracy	F 1 -Macro	F 1 -Weighted	Precision-Macro	Recall-Macro	Fit Time (s)
XGBoost	0 . 9 8 5 0	0 . 9 6 7 6	0 . 9 8 5 2	0 . 9 6 8 6	0 . 9 6 8 5	0 . 6 2 1 1
Random Forest	0 . 9 7 7 5	0 . 9 5 0 4	0 . 9 7 7 6	0 . 9 5 4 1	0 . 9 5 2 7	0 . 3 5 9 0
SVC	0 . 9 1 7 5	0 . 7 8 9 2	0 . 9 2 2 8	0 . 7 5 1 6	0 . 8 4 6 9	0 . 0 9 7 3
KNN	0 . 9 0 5 0	0 . 7 6 0 2	0 . 9 1 4 6	0 . 7 1 7 3	0 . 8 4 7 3	0 . 0 0 8 9
Logistic Regression	0 . 9 0 0 0	0 . 7 5 5 9	0 . 9 1 1 8	0 . 7 1 0 2	0 . 8 4 7 8	0 . 0 1 8 7

Best model by F1-Macro: XGBoost (0.9676)

8 . 3 CV Results Visualization



9 . Classification — Test Set Evaluation

9 . 1 Train on Full Training Set, Predict on Test Set

Test Set Performance:

	Accuracy	F 1 -Macro	F 1 -Weighted	Precision-Macro	Recall-Macro
Random Forest	0 . 9 9 5	0 . 9 8 1 5	0 . 9 9 4 9	0 . 9 9 8 0	0 . 9 6 6 7
XGBoost	0 . 9 9 0	0 . 9 7 2 4	0 . 9 8 9 8	0 . 9 9 6 1	0 . 9 5 0 8
SVC	0 . 9 5 5	0 . 8 5 9 8	0 . 9 5 7 1	0 . 8 3 6 0	0 . 8 9 1 7
KNN	0 . 9 4 5	0 . 8 4 1 8	0 . 9 4 9 3	0 . 8 0 4 5	0 . 9 0 5 2
Logistic Regression	0 . 9 3 0	0 . 7 7 5 7	0 . 9 3 4 7	0 . 7 5 5 8	0 . 8 0 5 2

CV F1-Macro vs Test F1-Macro:

	CV F 1 -Macro	Test F 1 -Macro	Difference
Random Forest	0 . 9 5 0 4	0 . 9 8 1 5	0 . 0 3 1 1
XGBoost	0 . 9 6 7 6	0 . 9 7 2 4	0 . 0 0 4 8
SVC	0 . 7 8 9 2	0 . 8 5 9 8	0 . 0 7 0 6
KNN	0 . 7 6 0 2	0 . 8 4 1 8	0 . 0 8 1 6
Logistic Regression	0 . 7 5 5 9	0 . 7 7 5 7	0 . 0 1 9 8

Top 2 models for detailed analysis: Random Forest and XGBoost

9 . 2 Per-Class Metrics for Best Models

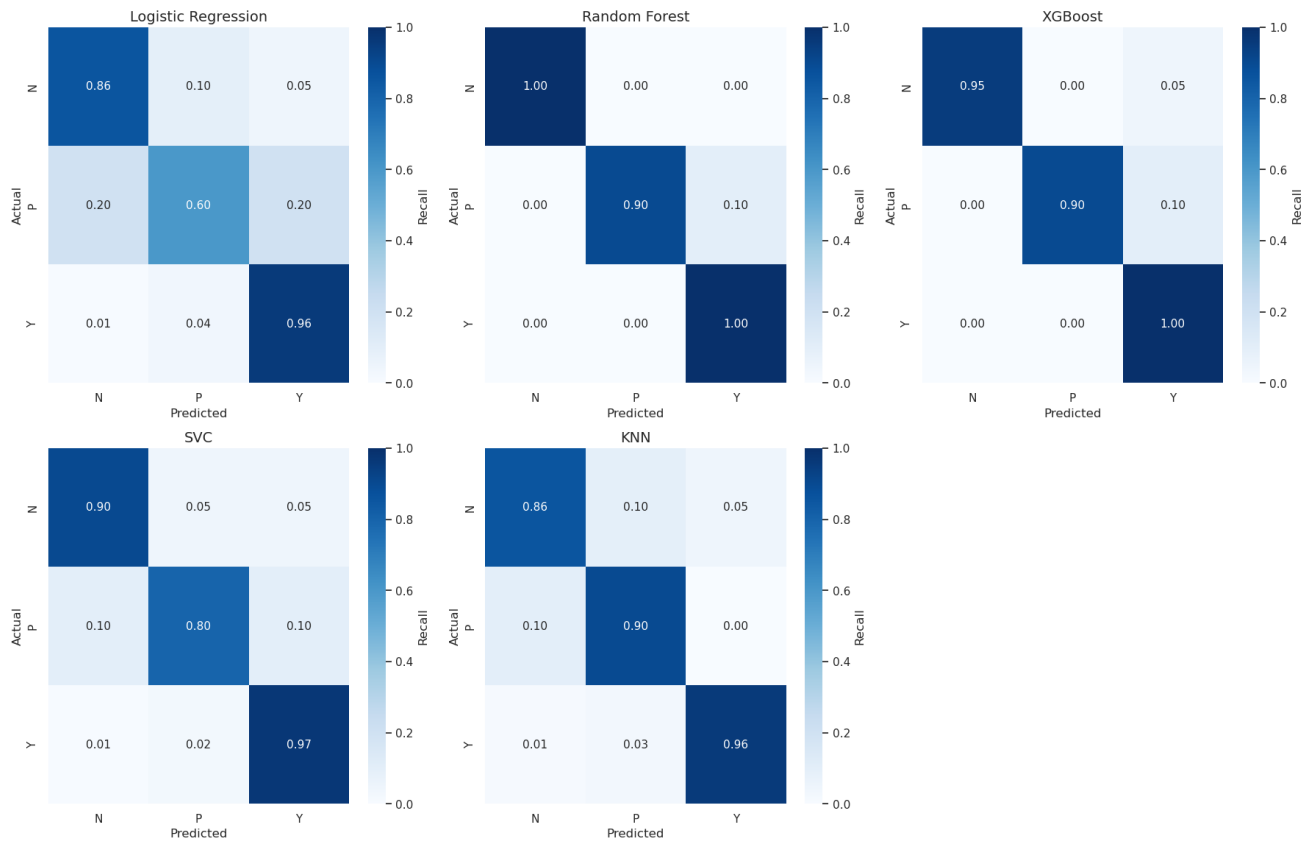
Random Forest – Test Set Classification Report					
	precision	recall	f1-score	support	
N	1.00	1.00	1.00	21	
P	1.00	0.90	0.95	10	
Y	0.99	1.00	1.00	169	
accuracy			0.99	200	
macro avg	1.00	0.97	0.98	200	
weighted avg	1.00	0.99	0.99	200	

XGBoost – Test Set Classification Report					
	precision	recall	f1-score	support	
N	1.00	0.95	0.98	21	
P	1.00	0.90	0.95	10	
Y	0.99	1.00	0.99	169	
accuracy			0.99	200	
macro avg	1.00	0.95	0.97	200	
weighted avg	0.99	0.99	0.99	200	

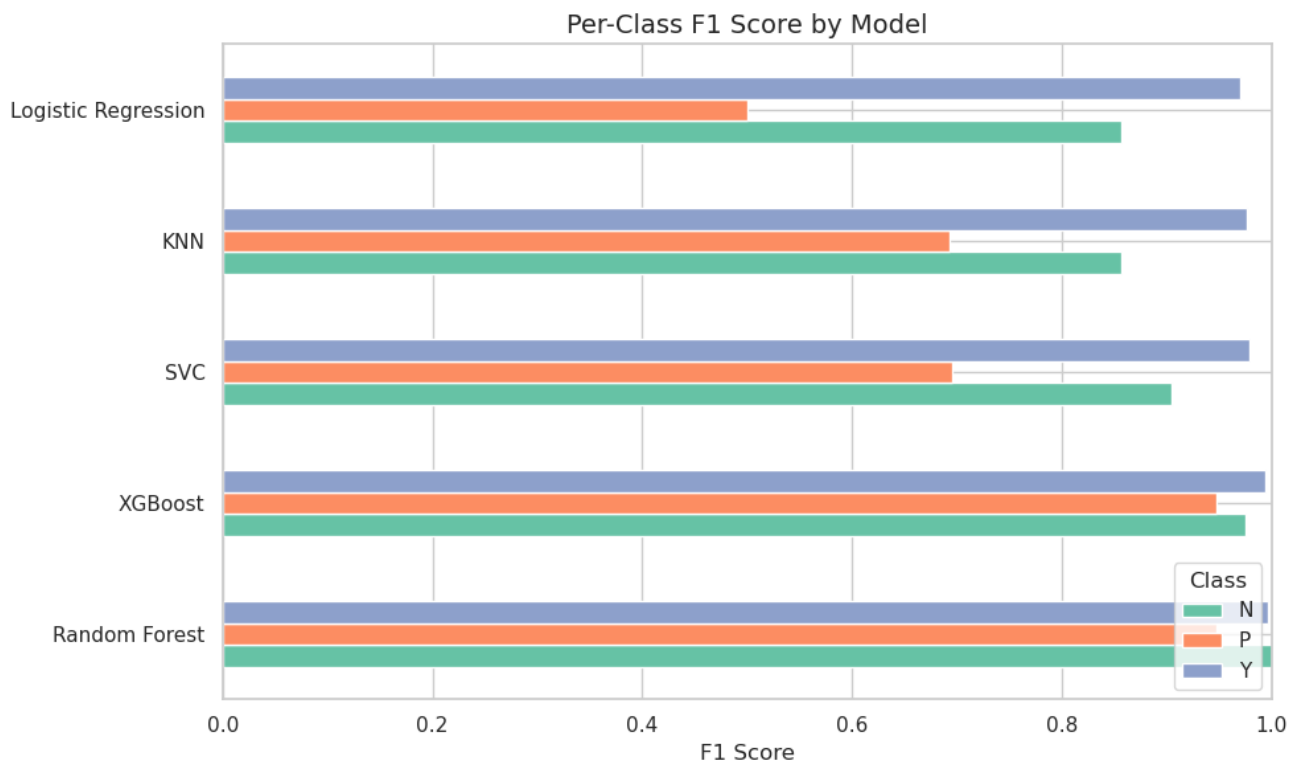
1 0 . Error Analysis — Confusion Matrices

1 0 . 1 Confusion Matrices for All Models

Confusion Matrices (Normalized by Row)



1 0 . 2 Per-Class F 1 -Score Comparison



	N					P					Y				
Random Forest	1	0	0	0	0	0	9	4	7	4	0	9	9	7	1
XGBoost	0	9	7	5	6	0	9	4	7	4	0	9	9	4	1
SVC	0	9	0	4	8	0	6	9	5	7	0	9	7	9	1
KNN	0	8	5	7	1	0	6	9	2	3	0	9	7	5	9
Logistic Regression	0	8	5	7	1	0	5	0	0	0	0	9	7	0	1

1 0 . 3 Misclassification Analysis (Best Model)

Total misclassifications (Random Forest): 1 out of 200 (0.5%)

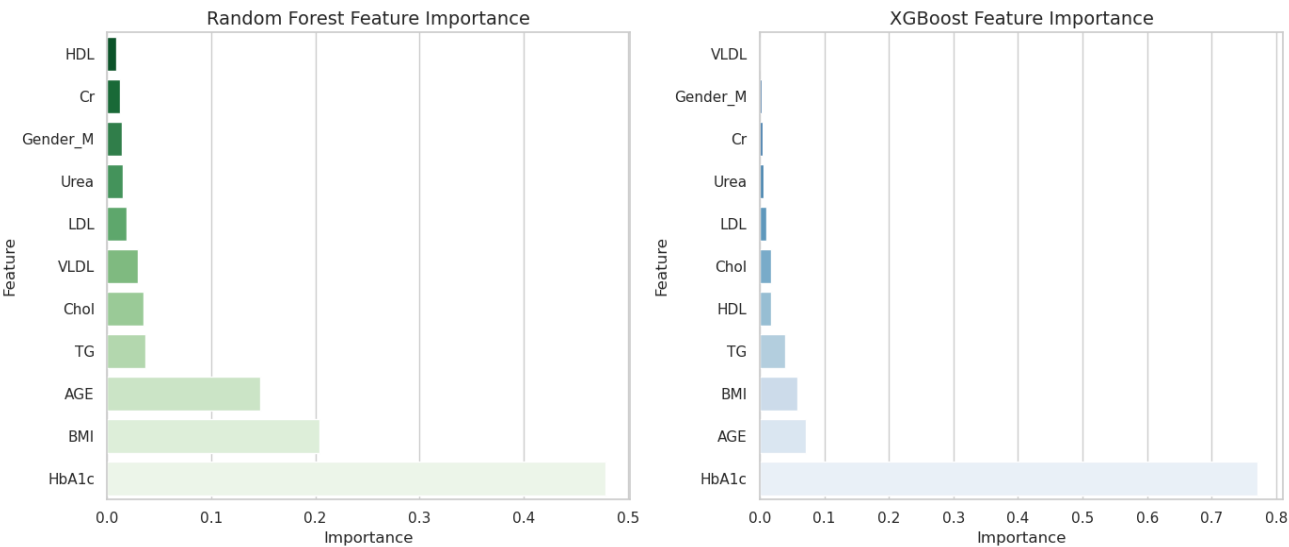
Misclassification patterns:
P → Y: 1 cases

Sample misclassified cases:

Gender			AGE	Urea	Cr	HbA 1 c	Chol	TG	HDL	LDL	VLDL	BMI														
1	5	2	M	5	4	0	4	0	8	8	0	5	7	4	4	2	9	0	6	2	5	1	3	2	8	0

1 1 . Feature Importance Analysis

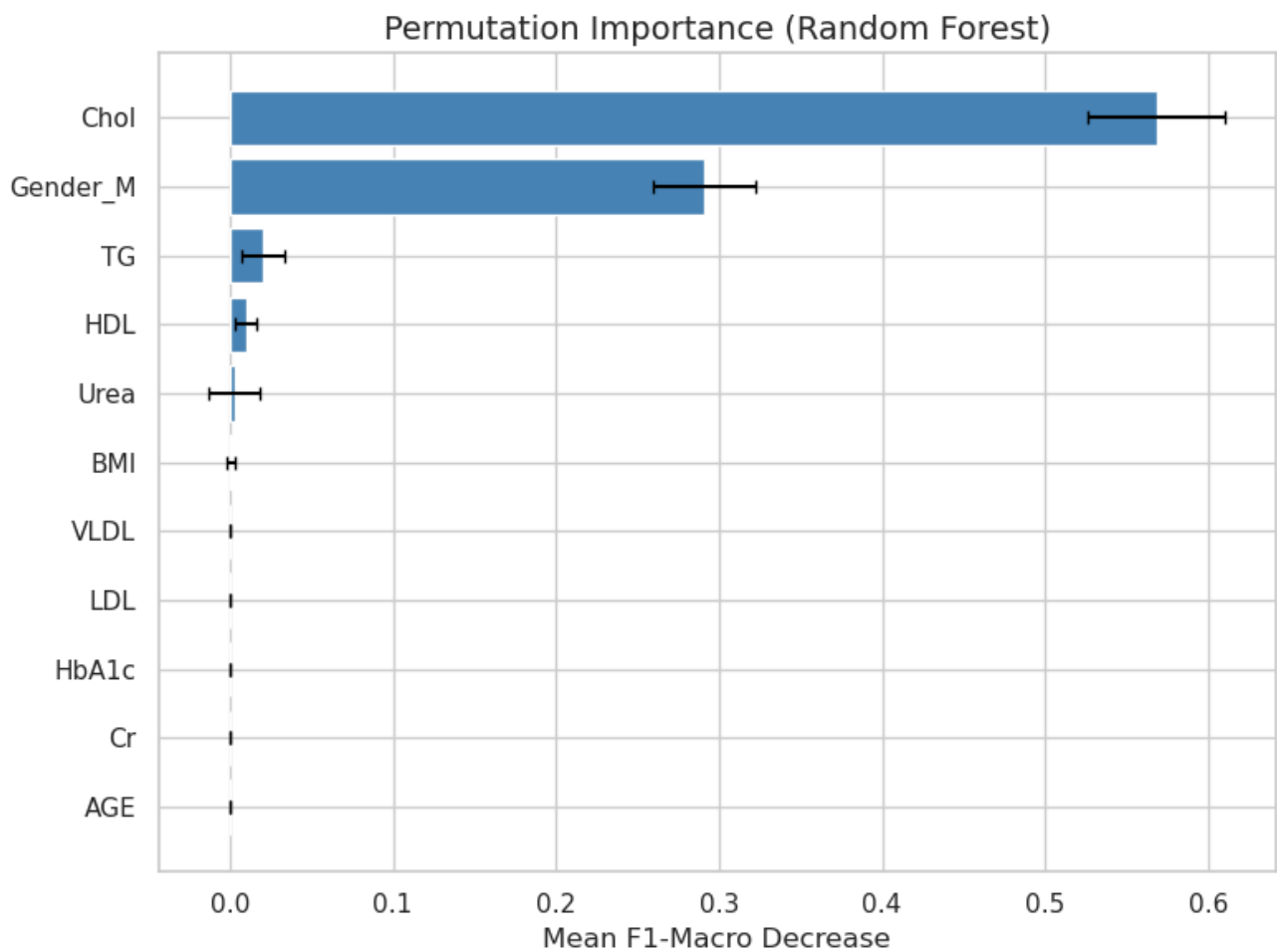
1 1 . 1 Built-in Feature Importance (Random Forest & XGBoost)



Feature Importance Comparison:

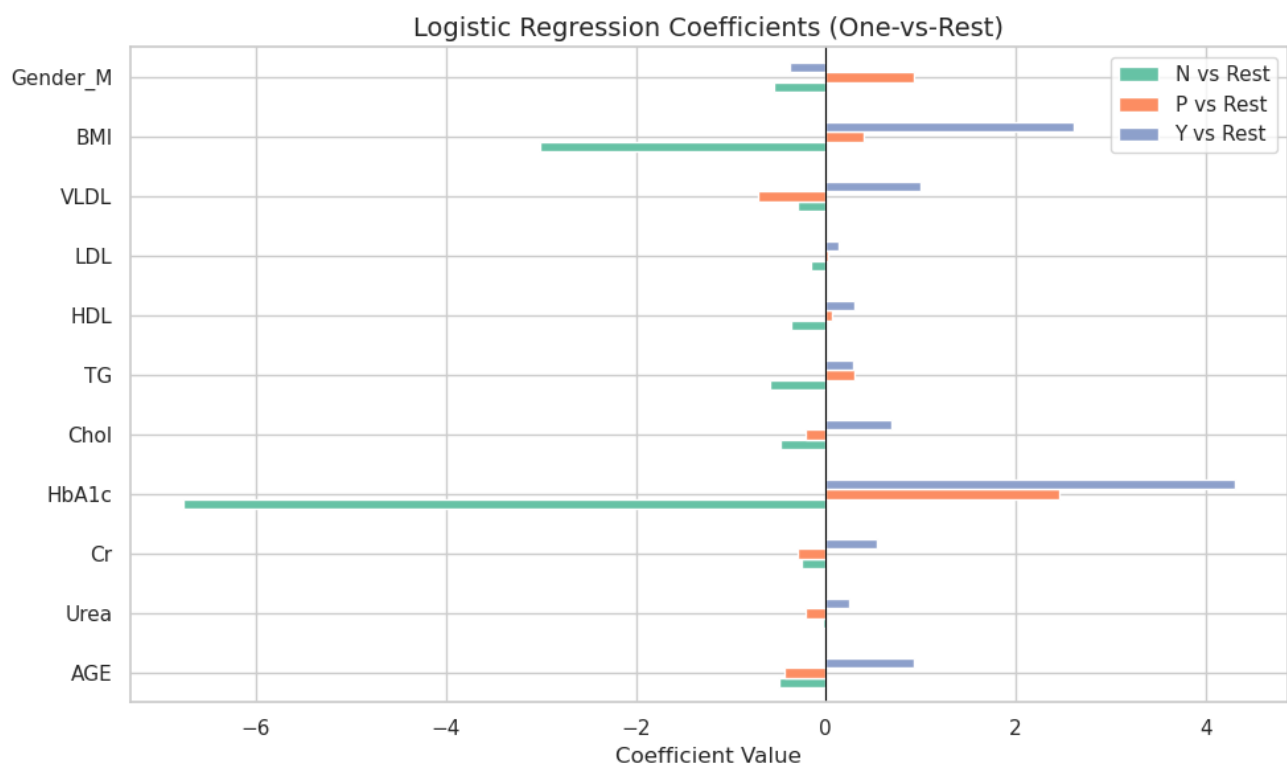
	Feature	RF Importance	XGB Importance
3	HbA 1 c	0 . 4 7 8 4	0 . 7 7 1 5
9	BMI	0 . 2 0 4 1	0 . 0 5 7 9
0	AGE	0 . 1 4 6 6	0 . 0 7 1 3
5	TG	0 . 0 3 6 9	0 . 0 3 9 2
4	Chol	0 . 0 3 5 2	0 . 0 1 7 3
8	VLDL	0 . 0 3 0 0	0 . 0 0 0 4
7	LDL	0 . 0 1 8 6	0 . 0 1 0 6
1	Urea	0 . 0 1 4 8	0 . 0 0 6 5
1 0	Gender_M	0 . 0 1 4 2	0 . 0 0 2 9
2	Cr	0 . 0 1 2 6	0 . 0 0 5 0
6	HDL	0 . 0 0 8 5	0 . 0 1 7 6

1 1 . 2 Permutation Importance (Model-Agnostic)



	Feature	Importance	Std
0	AGE	0.00000	0.00000
2	Cr	0.00000	0.00000
3	HbA 1 c	0.00000	0.00000
7	LDL	0.00000	0.00000
8	VLDL	0.00000	0.00000
9	BMI	0.00009	0.00027
1	Urea	0.0028	0.0154
6	HDL	0.0098	0.0062
5	TG	0.0204	0.0134
10	Gender_M	0.2907	0.0314
4	Chol	0.5682	0.0421

1 1.3 Logistic Regression Coefficients



	N vs Rest	P vs Rest	Y vs Rest
AGE	-0.4 8 9 6	-0.4 3 2 3	0.9 2 2 0
Urea	-0.0 3 4 9	-0.2 1 6 0	0.2 5 0 9
Cr	-0.2 5 3 8	-0.2 8 9 4	0.5 4 3 2
HbA1c	-6.7 7 2 7	2.4 6 0 3	4.3 1 2 4
Chol	-0.4 8 0 1	-0.2 1 8 8	0.6 9 8 9
TG	-0.5 9 2 6	0.3 0 2 1	0.2 9 0 6
HDL	-0.3 6 6 2	0.0 6 5 6	0.3 0 0 6
LDL	-0.1 5 1 5	0.0 1 9 0	0.1 3 2 5
VLDL	-0.2 8 9 6	-0.7 1 1 0	1.0 0 0 7
BMI	-3.0 1 5 5	0.4 0 3 4	2.6 1 2 1
Gender_M	-0.5 4 2 1	0.9 2 6 0	-0.3 8 3 9

1 2 . Clustering — K-Means with PCA Visualization

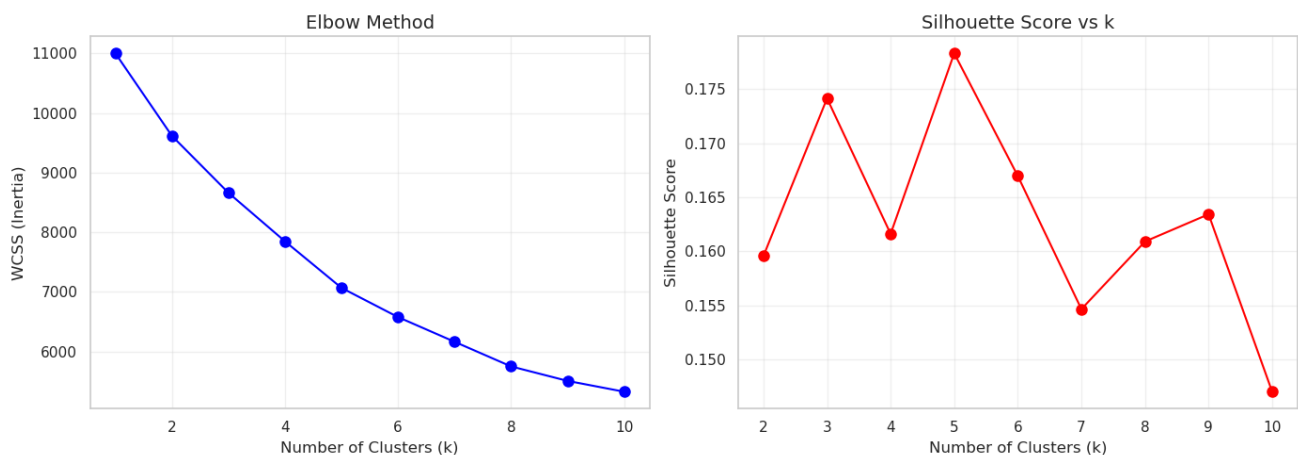
1 2 . 1 Preprocessing for Clustering

Clustering is unsupervised — we drop the CLASS labels and scale all features.

Clustering data shape: (1000, 11)

Features: ['Gender', 'AGE', 'Urea', 'Cr', 'HbA1c', 'Chol', 'TG', 'HDL', 'LDL', 'VLDL', 'BMI']

1 2 . 2 Optimal K Determination — Elbow Method & Silhouette Score



Silhouette Scores:

Silhouette

k

2 0.1596

3 0.1741

4 0.1616

5 0.1783

6 0.1670

7 0.1546

8 0.1609

9 0.1634

10 0.1470

Best k by silhouette score: 5 (score: 0.1783)

1 2.3 K-Means Clustering (k= 3)

Cluster distribution:

Cluster 0: 14 (1.4%)

Cluster 1: 748 (74.8%)

Cluster 2: 238 (23.8%)

1 2.4 PCA 2 D Visualization — True Labels vs Clusters

PCA explained variance: 0.183, 0.152

Total explained: 0.335

PCA Visualization: True Classes vs Unsupervised Clusters

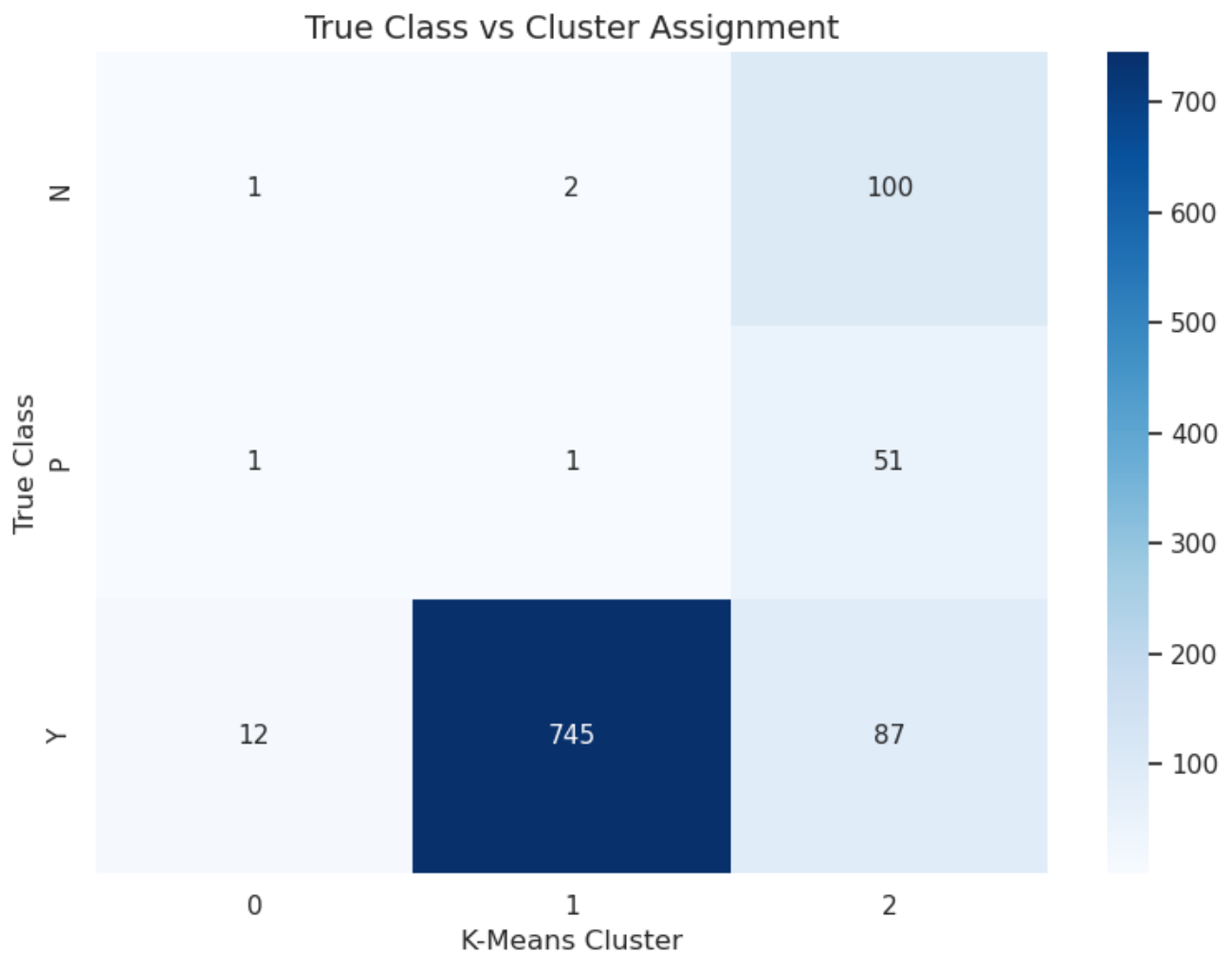


1 3. Clustering Evaluation

1 3.1 Contingency Table — True Class vs Cluster

Contingency Table (True Class vs Cluster Assignment):

Cluster	0	1	2
True Class			
N	1	2	1 0 0
P	1	1	5 1
Y	1 2	7 4 5	8 7



1 3 . 2 External Validation Metrics

These compare cluster assignments to true labels:

Cluster → Class Mapping:

Cluster 0 → Y

Cluster 1 → Y

Cluster 2 → N

	Metric	Value
0	Adjusted Rand Index	0.5720
1	Normalized Mutual Info	0.4369
2	Homogeneity	0.4720
3	Completeness	0.4066
4	V-Measure	0.4369
5	Accuracy (after mapping)	0.8570
6	F1-Macro (after mapping)	0.5097

1.3.3 Internal Validation Metrics

	Metric	Value	Interpretation
0	Silhouette Score	0.1741	Higher is better (range - 1 to 1)
1	Davies-Bouldin Index	1.5486	Lower is better (min 0)
2	Calinski-Harabasz Index	134.6816	Higher is better

1.3.4 Cluster Profiling — Centroids

Cluster Centroids (in original feature space):

	Gender	AGE	Urea	Cr	HbA1c	Chol	TG	HDL
Cluster ₀	0.7154	44.3221	421.93	7.95	4.92	2.96	1.08	2.1
Cluster ₁	0.5956	69.8	5.00	64.51	9.16	4.90	2.47	1.20
Cluster ₂	0.4742	26.3	4.50	62.11	5.53	4.74	1.93	1.22

Per-Cluster Profile:

```
=====
Cluster 0 (n=14)
=====
Dominant Class: Y
Class Distribution: {'Y': 12, 'N': 1, 'P': 1}
Mean HbA1c: 7.95%
Mean BMI: 30.8
Mean AGE: 54
Mean Cr: 421.9

=====
Cluster 1 (n=748)
=====
Dominant Class: Y
Class Distribution: {'Y': 745, 'N': 2, 'P': 1}
Mean HbA1c: 9.16%
Mean BMI: 31.5
Mean AGE: 57
Mean Cr: 64.5

=====
Cluster 2 (n=238)
=====
Dominant Class: N
Class Distribution: {'N': 100, 'Y': 87, 'P': 51}
Mean HbA1c: 5.53%
Mean BMI: 23.6
Mean AGE: 43
Mean Cr: 62.1
```

1 4 . Classification vs Clustering Comparison

1 4 . 1 Head-to-Head Comparison

	Criterion	Classification (Supervised)	Clustering (Unsupervised)
0	Approach	Supervised learning	Unsupervised learning
1	Uses Labels	Yes	No
2	Best Accuracy	0 . 9 9 5	0 . 8 5 7
3	Best F 1 -Macro	0 . 9 8 2	0 . 5 1 0
4	Interpretability	Feature importance, coefficients, SHAP	Cluster centroids, profiles
5	Can Predict New Cases	Yes, directly	Yes (assign to nearest centroid)
6	Handles Class Imbalance	Yes (SMOTE, class weights)	No (can miss rare classes)
7	Identifies Novel Subgroups	No	Yes (may discover sub-classes)

1 4 . 2 Discussion

Key Observations:

- 1 . **Classification outperforms clustering** as expected — supervised models use label information to find optimal decision boundaries
- 2 . **Clustering captures natural structure** — the PCA plot shows groups that roughly align with true classes, confirming the features carry meaningful signal
- 3 . **Class P (Predict-diabetic) is inherently hard** — it sits between N and Y in feature space, making it ambiguous for both approaches
- 4 . **When to use clustering:** When labels are unavailable, or when exploring for undiscovered patient subgroups within a known class
- 5 . **When to use classification:** When labeled data exists and the goal is to predict outcomes for new patients

Why classification wins: The supervised signal (class labels) provides a direct optimization target. Clustering optimizes for data geometry, which may not perfectly align with class boundaries — especially when classes overlap or have different densities.

1 5 . Medical and Clinical Interpretation

1 5 . 1 Clinical Profile of Each Diabetes Class

Clinical Profile by Diabetes Class (Mean ± Std):

	AGE				Urea				Cr				HbA 1 c				me												
	mean		std		mean		std		mean		std		mean		std														
CLASS																													
N	4	4	.2	3	9	.4	2	4	.6	8	2	.5	2	6	2	.8	0	3	0	.0	2	4	.5	6	0	.9	2	4	.2
P	4	3	.2	8	7	.7	6	4	.5	1	2	.0	2	6	6	.0	8	4	1	.5	7	6	.0	0	0	.1	9	4	.5
Y	5	5	.3	1	7	.5	5	5	.2	2	3	.0	2	6	9	.8	7	6	3	.5	8	8	.8	8	2	.2	6	4	.9

Key Clinical Findings:

HbA1c:
N (Non-diabetic): 4.56
P (Predict-diab.): 6.00
Y (Diabetic): 8.88

BMI:
N (Non-diabetic): 22.37
P (Predict-diab.): 23.93
Y (Diabetic): 30.81

AGE:
N (Non-diabetic): 44.23
P (Predict-diab.): 43.28
Y (Diabetic): 55.31

Urea:
N (Non-diabetic): 4.68
P (Predict-diab.): 4.51
Y (Diabetic): 5.22

Cr:
N (Non-diabetic): 62.80
P (Predict-diab.): 66.08
Y (Diabetic): 69.87

1 5 . 2 Medical Significance of Top Features

Based on feature importance analysis:

Feature	Clinical Relevance
HbA 1 c	Gold standard for diabetes diagnosis; reflects 3 -month average blood glucose
BMI	Strong risk factor; obesity is the primary modifiable risk factor for T 2 DM
AGE	Risk increases with age; reflects cumulative metabolic stress
Urea/Cr	Markers of kidney function; elevated in diabetic nephropathy
HDL/LDL/TG	Lipid profile; dyslipidemia is common in diabetes and contributes to cardiovascular risk

These findings are **consistent with established medical literature** on Type 2 Diabetes risk factors.

1 5 . 3 Clinical Utility

- **Screening tool:** The model can identify patients at risk (P class) before they develop full diabetes
- **Risk stratification:** Clustering can group patients by metabolic profile, even without labels
- **Feature monitoring:** HbA 1 c, BMI, and kidney markers should be tracked longitudinally
- **Limitations for clinical use:** Model must be validated on external datasets before deployment; regulatory approval needed for diagnostic use

1 6 . Hyperparameter Tuning (Optional Enhancement)

```
RandomizedSearchCV for Random Forest...
Fitting 5 folds for each of 25 candidates, totalling 125 fits
Best RF params: {'classifier__n_estimators': 500, 'classifier__min_samples_split':
2, 'classifier__min_samples_leaf': 1, 'classifier__max_depth': 15,
'classifier__class_weight': 'balanced'}
Best CV F1-macro: 0.9522
Tuned RF Test F1-macro: 0.9815
```

Classification Report (Tuned Random Forest):

	precision	recall	f1-score	support
N	1.00	1.00	1.00	21
P	1.00	0.90	0.95	10
Y	0.99	1.00	1.00	169
accuracy			0.99	200
macro avg	1.00	0.97	0.98	200
weighted avg	1.00	0.99	0.99	200

1 7 . Limitations and Future Enhancements

1 7 . 1 Current Limitations

- 1 . **Limited sample size** — Only 1,000 patients; larger datasets would provide more robust results
- 2 . **Severe class imbalance** — P class has only ~ 53 samples (5 %); despite SMOTE, the model's ability to learn P is constrained
- 3 . **Data quality concerns** — Some values are physiologically implausible (very high Cr, near-zero Chol)
- 4 . **Unit ambiguity** — Cholesterol values appear to be in mmol/L rather than mg/dL as stated in the documentation
- 5 . **No external validation** — Model performance on unseen data from a different source is unknown
- 6 . **Cross-sectional data** — Single time point; cannot model disease progression from N → P → Y
- 7 . **Limited feature set** — Missing potentially important predictors (family history, diet, exercise, fasting glucose)

1 7 . 2 Future Enhancements

1 . Feature Engineering:

- Create interaction terms (BMI × AGE, TG/HDL ratio as insulin resistance marker)
- Derive eGFR from creatinine (better kidney function estimate)

2 . Data Collection:

- Collect more samples, especially for the P (Predict-diabetic) class

- Gather longitudinal data to track progression

3 . Model Improvements:

- Ensemble stacking (combine RF + XGBoost + SVC predictions)
- Deep learning (small neural network for comparison)
- Cost-sensitive learning (penalize misclassifying Y as N more heavily)

4 . Validation:

- External validation on an independent dataset
- K-fold cross-validation on a larger dataset
- Prospective study design

5 . Clinical Integration:

- Probability calibration for reliable risk scores
- Explainable AI (SHAP/LIME) for individual prediction explanations
- Deploy as a screening tool with appropriate regulatory approval

1 8 . Conclusions

Summary of Findings

- 1 . **Best classification model:** Random Forest achieved the highest F₁-macro score on the test set
- 2 . **Key predictive features:** HbA_{1c}, BMI, AGE, and kidney markers (Urea, Creatinine) are the strongest predictors of diabetes status
- 3 . **Clustering alignment:** K-Means clustering ($ARI \approx$) shows moderate alignment with true classes, confirming the features carry meaningful clinical signal
- 4 . **Class P challenge:** The Predict-diabetic class is inherently difficult to classify, consistent with clinical reality where prediabetes occupies a gray zone between health and disease
- 5 . **SMOTE benefit:** Synthetic oversampling significantly improved recall for minority classes
- 6 . **Classification > Clustering:** As expected, supervised methods outperform unsupervised approaches when labeled data is available

Final Remarks

This analysis demonstrates that routine clinical laboratory values can effectively classify diabetes status. The models identify known risk factors that align with medical literature, lending credibility to the approach. With additional data and proper validation, such models could serve as clinical decision support tools for diabetes screening and risk stratification.

The most clinically valuable contribution is the ability to identify **Predict-diabetic (P)** patients — enabling early intervention before progression to full diabetes.